

Experiment 17 — Mobile Price Prediction

AIM

To implement a Mobile Price Prediction model using a suitable machine learning algorithm in Google Colab.

SOFTWARE USED

Google Colab

PROCEDURE

1. Open Google Colab and create a new notebook.
2. Import the required libraries.
3. Load the Mobile Price dataset.
4. Separate the input features and target variable.
5. Split the dataset into training and testing data.
6. Apply feature scaling to the input features.
7. Create a Random Forest Classifier.
8. Train the model using the training data.
9. Predict the mobile price range for the test data.
10. Calculate the accuracy of the model.
11. Display the confusion matrix.
12. Visualize the confusion matrix using a heatmap.

CODE

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
```

```
from sklearn.ensemble import RandomForestClassifier

from sklearn.metrics import accuracy_score, confusion_matrix

# Create mobile price dataset

data = {

    "battery_power": [842, 1021, 563, 615, 1821, 1859, 1253, 1675,
                      1500, 900, 1100, 1350, 1800, 700, 950, 1600,
                      1200, 1450, 1900, 800],

    "ram": [2549, 2631, 2603, 2769, 1411, 1067, 2303, 1087,
            1800, 1200, 2200, 2800, 3500, 1000, 1500, 3000,
            2000, 2500, 4000, 1100],

    "internal_memory": [7, 53, 41, 10, 44, 22, 49, 39,
                        32, 16, 64, 128, 256, 8, 16, 128,
                        32, 64, 256, 16],

    "battery_life": [19, 7, 9, 11, 8, 7, 20, 5,
                     15, 12, 18, 20, 22, 8, 10, 19,
                     15, 17, 24, 9],

    "screen_size": [5.0, 5.5, 4.7, 5.0, 6.2, 5.8, 5.1, 5.5,
                    5.7, 5.0, 6.0, 6.5, 6.7, 4.8, 5.2, 6.4,
                    5.8, 6.1, 6.8, 5.0],

    "price_range": [1, 1, 1, 1, 0, 0, 1, 0,
                    1, 0, 1, 2, 3, 0, 0, 2,
                    1, 2, 3, 0]

}

# Create DataFrame

df = pd.DataFrame(data)

# Display dataset

print("Dataset:")

print(df.head())
```

```
print("\nDataset Shape:", df.shape)

# Separate features and target
X = df.drop("price_range", axis=1)
y = df["price_range"]

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42,
    stratify=y
)

# Create Random Forest model
model = RandomForestClassifier(
    n_estimators=100,
    random_state=42
)

# Train model
model.fit(X_train, y_train)

# Predict
y_pred = model.predict(X_test)

# Calculate accuracy
accuracy = accuracy_score(y_test, y_pred)

print("\nAccuracy:", accuracy)

# Confusion Matrix
cm = confusion_matrix(y_test, y_pred)

print("\nConfusion Matrix:")

print(cm)
```

```

# Visualize confusion matrix
plt.figure(figsize=(6, 5))
sns.heatmap(
    cm,
    annot=True,
    fmt="d",
    cmap="Blues",
    xticklabels=["Low", "Medium", "High", "Very High"],
    yticklabels=["Low", "Medium", "High", "Very High"]
)
plt.xlabel("Predicted Price Range")
plt.ylabel("Actual Price Range")
plt.title("Mobile Price Prediction - Confusion Matrix")

plt.show()

```

OUTPUT

Dataset:

	battery_power	ram	internal_memory	battery_life	screen_size \
0	842	2549	7	19	5.0
1	1021	2631	53	7	5.5
2	563	2603	41	9	4.7
3	615	2769	10	11	5.0
4	1821	1411	44	8	6.2

	price_range
0	1
1	1
2	1
3	1
4	0

Dataset Shape: (20, 6)

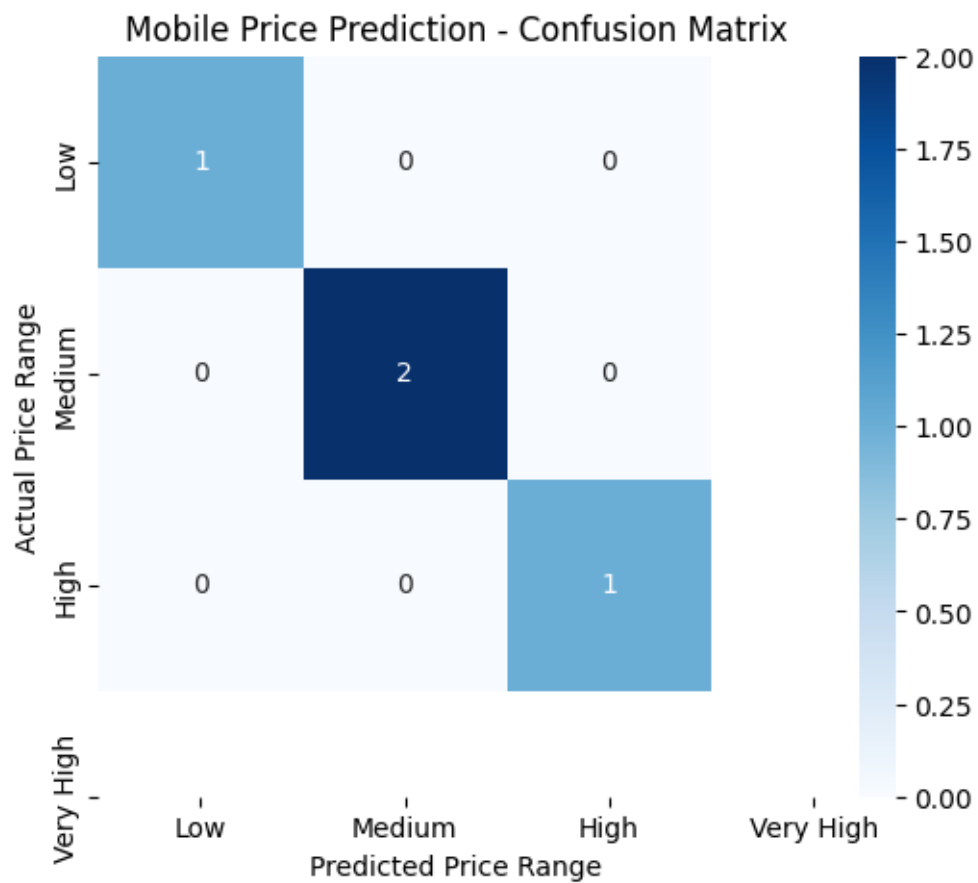
Accuracy: 1.0

Confusion Matrix:

```
[[1 0 0]
```

```
[0 2 0]
```

```
[0 0 1]]
```



RESULT

Thus, the Mobile Price Prediction model was successfully implemented using the Random Forest Classification algorithm in Google Colab, and mobile phones were classified into different price ranges.